TITLE: Device Assembly Process

1. MATERIAL REQUIRED:

Material requirement slips will be issued by unit testing person based on that parts are consumed.

2. TOOLS REQUIRED:

S. No.	Item name	UOM	Quantity
1.	Pipe Cutter	Nos	1
2.	Wire cutter	Nos	1
3.	Screwdriver	Nos	1
4.	Nose Plyer	Nos	1
5.	Cutting plyer	Nos	1
6.	Bosch Battery Screwdriver	Nos	1
7.	Teflon Tape	Nos	1

3. ACTIVITIES:

Filters and functional parts are assembled on to the device with necessary tools mentioned above.

1. Filter clam fixing
2. Pump & SV fixing
3. Wiring harness fixing
4. Filters & Membrane Assembly
5. Tubing's

Filter Clam Fixing:

1. Place the filters in their designated locations on the device.
2. Attach the clamp filters and secure them with screws or bolts.
3. Ensure clamps are tight enough to hold the filters firmly but not too tight to avoid damage.

Pump & SV (Solenoid Valve) Fixing:

1. Position the pump in its designated location.
2. Attach the mounting brackets to the pump.
3. Secure the pump to the device using screws or bolts.
4. Position the SV in its designated location.
5. Attach the mounting brackets to the SV.
6. Secure the SV to the device using screws or bolts.

Wiring Harness Fixing:

1. Route the wiring harness through the device, ensuring it does not interfere with other components.
2. Use cable ties to secure the wiring harness in place.
3. If necessary, use electrical tape to bundle wires together neatly.
4. Connect the wires to their respective components (e.g., pump, SV, sensors) as per the wiring diagram.
5. Ensure all connections are secure and properly insulated.

Filters & Membrane Assembly:

1. Insert the filters into their respective housings.
2. Ensure the filters are seated correctly and are in place to prevent transit damage.
3. Insert the RO membrane into its housing.
4. Ensure the membrane is properly seated and the housing is sealed tightly.
5. Use a filter wrench to tighten the filter housings if required.
6. Ensure all housing is sealed properly to prevent leaks.

Tubing:

1. Measure and cut the tubing to the required lengths.
2. Attach tubing to the filters, pump, SV, and other components using the appropriate connectors or fittings.
3. Ensure all connections are secure and free from leaks.
4. Use cable ties or clamps to secure the tubing in place, ensuring it does not interfere with other components.

4. SCRAP DISPOSAL INSTRUCTIONS:

1. Keep the polythene covers/packing carton in the respective bins only.
2. Keep the dust cap in the respective bins only.

5. SAFETY INSTRUCTIONS:

Keep the material away from the gang way.

Keep the Device cabinets covered with polythene cover to avoid dust & Scratches.

6. CHECKS:

1. Ensure properly that there should be no leakage @ 45 Psi.(as per BIS)
2. Ensure that the float switch is working properly.

PREPARED BY	REVIEWED BY	APPROVED BY	ISSUED BY

7. **RESPONSIBILITY:** Operator

Final QC Checks:

1. **Inspect All Connections:**
    - Ensure all filters, pump, SV, wiring harness, and tubing are properly secured and connected.
2. **Check for Leaks:**
    - Run Air through the system to check for any leaks in the tubing or filter housings.
3. **Test Functionality:**
    - Power on the device and test the functionality of all components (e.g., pump operation, SV opening/closing, Air flow through filters with leakage test).
4. **Final Inspection:**
    - QC Person perform a final inspection to ensure everything is assembled correctly and the device is functioning as expected.